Robot Dynamics Midterm 1

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Duration: 1h 15min
Permitted Aids: The exam is open book, which means you can use the script, slides, exercises, etc; The use of internet (besides for licenses) is forbidden; no communication among students during the test is allowed.

1 Instructions

1. Download the ZIP file `RobotDynamics_Quiz1_2025.zip` from Moodle. Extract all contents of this file into a new folder and set MATLAB's[1] current path to this folder.
2. Run `init_workspace` in the Matlab command line.
3. All problem files that you need to complete are located in the `problems` folder.
4. Run `evaluate_problems` to check if your functions run. This script does not test for correctness. You will get 0 points if a function does not run (e.g., for syntax errors).
5. When the time is up, zip the entire folder and name it `ETHStudentID_StudentName.zip`, for example, `01234567_JohnDoe.zip`. Submit this zip-file through Moodle under **Midterm Exam 1 Submission**. You should receive a confirmation email.
6. If the previous step did not succeed, you can email your file to `robotdynamics@leggedrobotics.com` from your ETH email address with the subject line `[RobotDynamics] ETHStudentID - StudentName`
7. **Important:**
   (a) Implementations outside the provided templates will not be graded and receive 0 points.
   (b) The use of functions in MATLAB Toolboxes outside of the base MATLAB installation is not allowed.
   (c) Helper functions included in the `solutions/pcode` directory are specifically for Question 4. Using these functions in the solutions for other questions is prohibited and will receive 0 points.

[1] Online version of MATLAB at `https://matlab.mathworks.com/`

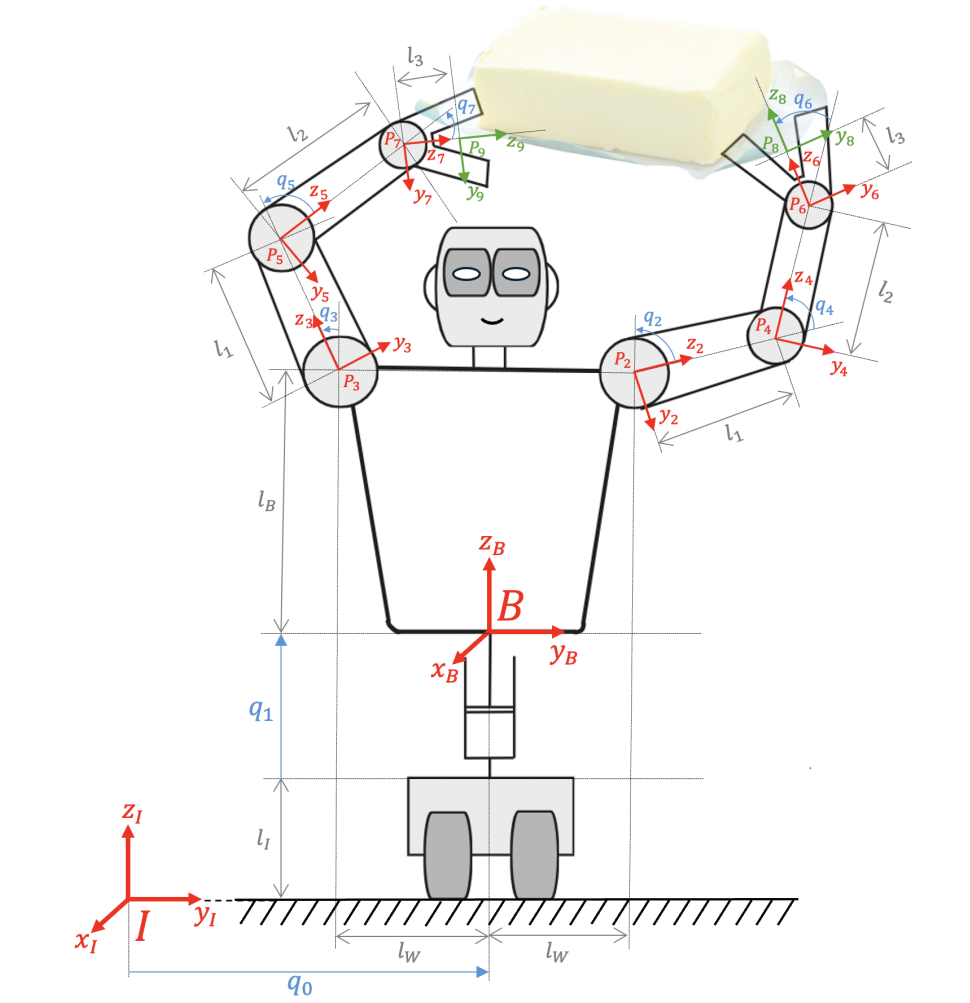


Figure 1: Schematic of a humanoid robot with two three degrees of freedom arms featuring unactuated grippers and a height-adjustable base. The base can move linearly along the y_I and z_I directions. All revolute joints on the arms rotate around the positive x_I axis. The x axis of frames $\{P_2\}, \{P_3\}, \ldots, \{P_7\}$ is parallel to the x_I axis.

2 Questions

The eccentric scientist, Dr. Richard, often faced a recurring problem — the butter was always just out of reach on the dining table. Determined to fix this, he designed a small humanoid robot capable of moving around and passing condiments. However, before he could develop its controller, he was called away on an intergalactic mission, leaving the task to you.

In this quiz, you will model the forward and differential kinematics of the humanoid robot shown in Fig. 1. The robot has two arms with 3 Degrees-of-Freedom (DoF) each, connected to a mobile, height-adjustable base. Each arm has an unactuated gripper.

The base can move linearly along the y_I and z_I axes . The reference frame attached to the base is denoted as $\{B\}$. The arms are composed of three links each. The reference frames attached to each link are denoted as $\{P_2\}, \{P_3\}, \ldots, \{P_9\}$.

The generalized coordinates are defined as

$$\boldsymbol{q} = \begin{bmatrix} q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7 \end{bmatrix}^T \in \mathbb{R}^8 \, . \tag{1}$$

Clarification: The angles $q_2, q_3, \cdots, q_7$ are measured along the positive x-axis using right-hand thumb rule. For the state of the scene shown in Fig. 1, q_2, q_5 and q_7 would have negative values, and q_3, q_4 and q_6 would have positive values.

In the following questions, all required parameters are passed to your functions in a structure called `params`. You can access it as follows:

```
lI = params.lI;
lW = params.lW;
lB = params.lB;
l1 = params.l1;
l2 = params.l2;
l3 = params.l3;
```

Question 1. 6 P.

Find the homogeneous transformation between the inertial frame $\{I\}$ and the left gripper frame $\{P_8\}$, i.e., the matrix $\mathbf{T}_{IP_8}$ as a function of the generalized coordinates $\boldsymbol{q}$.

You should implement your solution in the function `jointToEEPose.m`

Question 2. 4 P.

Compute the position Jacobian of the left gripper ${}_I\mathbf{J}_{IP_8,P} \in \mathbb{R}^{3\times 8}$, that fulfills:

$${}_I\boldsymbol{v}_{IP_8} = {}_I\mathbf{J}_{IP_8,P}(\boldsymbol{q})\dot{\boldsymbol{q}}, \tag{2}$$

where ${}_I\boldsymbol{v}_{IP_8} \in \mathbb{R}^3$ is the linear velocity of point P_8 (the left gripper frame origin) with respect to a fixed point expressed in frame $\{I\}$.
You should implement your solution in the function `jointToPositionJacobian.m`

Question 3. 2 P.

Compute the rotation Jacobian of the left gripper ${}_I\mathbf{J}_{IP_8,R} \in \mathbb{R}^{3\times 8}$, that fulfills:

$${}_I\boldsymbol{\omega}_{IP_8} = {}_I\mathbf{J}_{IP_8,R}(\boldsymbol{q})\dot{\boldsymbol{q}}, \tag{3}$$

where ${}_I\boldsymbol{\omega}_{IP_8} \in \mathbb{R}^3$ is the angular velocity of frame $\{P_8\}$ with respect to the inertial frame $\{I\}$, expressed in frame $\{I\}$.
You should implement your solution in the function `jointToRotationJacobian.m`

Question 4. 3 P.

Dr. Richard enjoys his pancakes only when the maple syrup is poured just right, in a perfectly smooth, preplanned pattern. To make breakfast as precise as his science, he designed a trajectory for the robot's arm to follow while pouring the syrup.

Implement a kinematic trajectory tracking controller that is able to track desired poses for the left ${}_I\boldsymbol{p}^*_{IP_8} = [{}_Iy^*_{IP_8}, {}_Iz^*_{IP_8}, {}_I\phi^*_{IP_8}]^T$ and right ${}_I\boldsymbol{p}^*_{IP_9} = [{}_Iy^*_{IP_9}, {}_Iz^*_{IP_9}, {}_I\phi^*_{IP_9}]^T$ grippers as a function of the generalized coordinates $\boldsymbol{q}$.

The desired gripper poses ${}_I\boldsymbol{p}^*_{IP_8}$ and ${}_I\boldsymbol{p}^*_{IP_9}$, as well as the current joint angles $\boldsymbol{q}$, are passed as inputs to your MATLAB function.

For this question, we provide the following:

- A function to calculate the current left and right gripper 2D poses in the plane ${}_I\boldsymbol{p}_{IP_8}$ and ${}_I\boldsymbol{p}_{IP_9}$ given the current generalized coordinates $\boldsymbol{q}$. You can call it with: `[p_I8_I, p_I9_I] = jointTo2DPose_solution(q, params);`
- The analytical Jacobians of the left and right grippers $\mathbf{J}_{P_8} \in \mathbb{R}^{3\times 8}$ and $\mathbf{J}_{P_9} \in \mathbb{R}^{3\times 8}$ that fulfill:

$$ {}_I\boldsymbol{w}_{P_8} \;=\; \mathbf{J}_{P_8}(q)\,\dot{\mathbf{q}} \qquad \text{and} \qquad {}_I\boldsymbol{w}_{P_9} \;=\; \mathbf{J}_{P_9}(q)\,\dot{\mathbf{q}}. \tag{4} $$

  You can call it with: `[J_I8_2d, J_I9_2d] = jointTo2DJacobian_solution(q, params);`
- A function to compute the damped pseudo-inverse of a matrix `A`. You can call it with: `A_pinv = pseudoInverseMat_solution(A, lambda)`.

Note: The correct way of parsing a tuple from a function in Matlab is as follows: `[a, b] = functionThatReturnsTuple(q, params);`

You should implement your solution in the function `kinematicTrajectoryControl.m`.

Your implementation is judged based on how well the robot's left and right grippers track a predefined trajectory. You can visualize your implementation of the tracking controller by running `trajectory_control_loop`. You may also check the tracking of the solution by setting `use_solution` to 1 in the file `trajectory_control_loop.m`